

Data Evidence Appendix

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Source artifact: artifacts/institutional/idea1-data-evidence-appendix.md

Idea 1 — BTC Vol Desk Data & Evidence Appendix

Purpose: Establish the evidence spine for a BTC ETF / Deribit / CME / OTC volatility intermediation desk before writing investor narrative.

Standard: Every data source is classified as verified, credentialed/paid, or hypothesis. Screen-only data is not treated as executable.

1. Source classification

A. Deribit BTC options — verified public API

Status: Verified public API access.

Official docs / endpoints:

- OpenAPI spec: ``https://docs.deribit.com/specifications/deribit_openapi.json``
- Instruments: ``GET https://www.deribit.com/api/v2/public/get_instruments?currency=BTC&kind=option&expired=false``
- Book summary: ``GET https://www.deribit.com/api/v2/public/get_book_summary_by_currency?currency=BTC&kind=option``
- Order book: ``GET https://www.deribit.com/api/v2/public/get_order_book?instrument_name=BTC-...&depth=1``
- Historical volatility: ``GET https://www.deribit.com/api/v2/public/get_historical_volatility?currency=BTC``

Live verification from May 14, 2026 pass:

- Active BTC option instruments: ~936
- Expiries observed: 12
- Near-ATM IV samples:
 - 15MAY26: ~31.1%
 - 22MAY26: ~35.1%
 - 29MAY26: ~36.5%
 - 26JUN26: ~38.1%
 - 25SEP26: ~40.8%

Useful fields:

- Metadata: ``instrument_name`, `instrument_id`, `expiration_timestamp`, `strike`, `option_type`, `state`, `is_active`, `contract_size`, `settlement_currency`, `quote_currency`, `price_index``
- Market snapshot: ``bid_price`, `ask_price`, `mid_price`, `mark_price`, `mark_iv`, `underlying_price`, `underlying_index`, `interest_rate`, `open_interest`, `volume`, `volume_usd`, `estimated_delivery_price``

- Top-of-book: `best\_bid\_price`, `best\_bid\_amount`, `best\_ask\_price`, `best\_ask\_amount`, `bid\_iv`, `ask\_iv`, `mark\_iv`, `greeks.delta`, `greeks.gamma`, `greeks.vega`, `greeks.theta`, `greeks.rho`
- Historical vol: live payload returns `[timestamp, value]` pairs despite docs schema implying `timestamp` / `value` objects.

Caveats:

- Deribit prices are crypto-native and must be normalized into USD/BTC-equivalent terms.
- Use WebSocket subscriptions for real-time production instead of polling every instrument.
- Rate limits are credit-based; `get\_instruments` has stricter limits than normal endpoints.

B. IBIT / listed ETF options — verified semi-official + official supporting sources

Nasdaq option chain

Status: Verified public JSON, but undocumented/semi-official rather than guaranteed production API.

Endpoint:

- `https://api.nasdaq.com/api/quote/IBIT/option-chain?assetclass=etf`
- Greeks endpoint: `https://api.nasdaq.com/api/quote/IBIT/option-chain/greeks?assetclass=etf`

Live verification from May 14, 2026 pass:

- IBIT last shown: `\$46.17 as of May 14, 2026`
- `totalRecord`: 412
- Expiries returned included May 15, May 18, May 20, May 22, May 26, May 27, May 29.
- Retrieved chain sample showed:
- call OI: ~1.12M contracts
- put OI: ~640k contracts
- call volume: ~336k
- put volume: ~109k

Useful fields:

- Chain fields: `expiryDate`, `strike`, `c\_Last`, `c\_Bid`, `c\_Ask`, `c\_Volume`, `c\_Openinterest`, `p\_Last`, `p\_Bid`, `p\_Ask`, `p\_Volume`, `p\_Openinterest`
- Greeks endpoint: `cDelta`, `cGamma`, `cRho`, `cTheta`, `cVega`, `cIV`, `pDelta`, `pGamma`, `pRho`, `pTheta`, `pVega`, `pIV`

Caveats:

- Not documented as a stable API contract.
- `table.asOf` observed as null.
- Display-oriented records, not OPRA-grade records.
- No guaranteed history, timestamps, quote conditions, NBBO provenance, or depth.
- Greeks methodology/source not specified.

OCC series search

Status: Official public source for listed series and open interest; not a quote/IV source.

Endpoint tested:

- ``https://marketdata.theocc.com/series-search?symbolType=U&symbol=IBIT``

Useful fields:

- Product symbol
- Expiration year/month/day
- Strike
- Call/put indicator
- Call OI
- Put OI
- Position limit
- Exchanges where product trades

Caveats:

- No bid/ask/last/volume/IV/Greeks.
- Useful as official series/OI reference, not as tradeable market data.

C. IBIT NAV / BTC-per-share normalization — verified official BlackRock/iShares source

Status: Verified official public iShares data.

Sources:

- Product page: ``https://www.ishares.com/us/products/333011/ishares-bitcoin-trust-etf``
- Holdings CSV:
``https://www.ishares.com/us/products/333011/fund/1467271812596.ajax?fileType=csv&fileName=IBIT_holdings&dataType=fund``

Observed official fields from May 13/14 pass:

- NAV: ``$45.05`` as of May 13, 2026
- Sponsor fee: ``0.25%``
- Net assets: ``$65,235,245,502``
- Shares outstanding: ``1,448,200,000``
- BTC quantity in holdings CSV: ``817,092.85700 BTC``
- Benchmark index: CME CF Bitcoin Reference Rate - New York Variant
- Bloomberg index ticker: BRRNY

Normalization calculation:

- $\text{BTC/share} = \text{BTC quantity} / \text{shares outstanding}$
- Using observed values: ``817,092.857 / 1,448,200,000 = 0.000564212717166 BTC/share``
- Shares per BTC $\approx$ ``1,772.38117748``

Use in vol monitor:

- Convert IBIT option strikes to BTC-equivalent strikes via BTC/share.
- Convert IBIT option premiums to BTC-equivalent or USD-per-BTC premium.
- Compare IV/Greeks only after wrapper and forward assumptions are normalized.

Caveats:

- Public holdings/NAV data is generally as-of/T-1, not real-time.
- Holdings CSV warns market value/weight/notional may use third-party vendor pricing.
- Use official quantity and shares outstanding for BTC/share; avoid inferring from market value.

D. CME BTC futures/options — verified official docs, credentialed for production data

Status: Official public web pages are reference-only; automated production data requires credentials/licensing/API/vendor.

Public product pages:

- Bitcoin futures: ``https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.html``
- Bitcoin futures quotes: ``https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.quotes.html``
- Bitcoin futures specs: ``https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.contractSpecs.html``
- Bitcoin options: ``https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.quotes.options.html``
- Bitcoin options specs: ``https://www.cmegroup.com/markets/cryptocurrencies/bitcoin/bitcoin.contractSpecs.options.html``

Automation caveat:

- CME public website returned HTTP 403 to scripted access and explicitly blocks scraping under its website data terms.
- Treat public pages as human reference/display only.

Official CME API/feed paths:

- Real-Time Futures and Options WebSocket API docs: ``https://cmegroupclientsite.atlassian.net/wiki/spaces/EPICSANDBOX/pages/457414253/CME+Market+Data+Over+WebSocket+API``
- Message spec: ``https://cmegroupclientsite.atlassian.net/wiki/display/EPICSANDBOX/CME+Market+Data+Over+WebSocket+API+-+Message+Specification``
- Production WebSocket: ``wss://markets.api.cmegroup.com/marketdatastream/v1``
- OAuth token production: ``https://auth.cmegroup.com/as/token.oauth2``
- MDP 3.0 market data docs: ``https://cmegroupclientsite.atlassian.net/wiki/display/EPICSANDBOX/CME+MDP+3.0+Market+Data``
- DataMine: ``https://www.cmegroup.com/datamine.html``
- DataMine API: ``https://www.cmegroup.com/datamine/datamine-api.html``

Useful official data classes:

- Top of book: bid/ask price, quantity, order count, timestamps, instrument metadata.
- Trade summary: trade price/quantity, aggressor side, timestamps, instrument metadata.
- Statistics: cleared volume, open interest, settlement price, settlement flags/date/timestamp.
- Historical DataMine: end-of-market summary, BBO, volume/OI.
- CME options analytics product appears to include Greeks and implied volatility, but requires credentials/entitlements.

Practical path:

- MVP: use broker/vendor data for CME BTC futures/options if available.
- Institutional: CME WebSocket API + DataMine historical after entitlements.
- Low-latency institutional: MDP 3.0 direct feed only if justified.

E. OTC / RFQ layer — not yet verified; must be quote-verified

Status: Hypothesis until live counterparties provide indicative quotes or executed RFQs.

Target sources:

- Crypto OTC desks
- Options market makers
- Prime brokers
- FCMs / institutional crypto brokers
- RFQ venues
- Structured-product desks

Required fields:

- Indicative bid/ask for standard structures
- Minimum ticket size
- Max executable size
- Counterparty credit terms
- Collateral requirements
- Settlement method
- Legal documents required
- Quote expiry
- Venue hedge assumptions
- Fees/commission/spread

Confidence labels:

- Hypothesis: based on expected market behavior only.
- Screen-only: based on listed market data only.
- Quote-verified: at least one real indicative quote from a counterparty.
- Trade-verified: actual executed trade or executable firm market.

2. Normalized BTC-equivalent schema

Every option record should be transformed into a common schema before comparison.

Required normalized fields

- `source`: Deribit / Nasdaq / OCC / CME / OTC / vendor
- `venue`: Deribit / ETF-listed / CME / OTC
- `instrument\_id`
- `instrument\_name`
- `underlying\_wrapper`: BTC index / IBIT ETF / CME futures / OTC reference
- `underlying\_price\_native`
- `btc\_reference\_price`
- `expiry\_date`
- `days\_to\_expiry`
- `strike\_native`
- `strike\_btc\_equivalent`
- `option\_type`: call / put
- `bid\_native`
- `ask\_native`
- `mid\_native`
- `mark\_native`
- `bid\_usd\_equivalent`
- `ask\_usd\_equivalent`
- `mid\_usd\_equivalent`
- `bid\_btc\_equivalent`
- `ask\_btc\_equivalent`
- `iv\_bid`
- `iv\_ask`
- `iv\_mid`
- `iv\_mark`
- `delta`
- `gamma`
- `vega`
- `theta`
- `rho`
- `open\_interest\_native`
- `volume\_native`
- `bid\_size`
- `ask\_size`
- `bid\_ask\_width\_pct`
- `forward\_price\_assumption`
- `rate\_assumption`

- `funding\_or\_basis\_assumption`
- `margin\_model`
- `collateral\_currency`
- `fee\_assumption`
- `slippage\_assumption`
- `data\_timestamp`
- `source\_confidence`: verified_public / semi_official / credentialed / paid_vendor / hypothesis
- `execution\_confidence`: screen_only / quote_verified / trade_verified

Wrapper-specific normalization

Deribit:

- Native quote is often in BTC terms for options.
- Use Deribit `underlying\_price` / `estimated\_delivery\_price` to convert to USD-equivalent.
- Confirm exact option contract settlement and PnL convention per instrument metadata.

IBIT:

- Compute BTC/share from official iShares holdings.
- Convert IBIT strike to BTC-equivalent: `strike\_native / BTC\_per\_share` only after checking whether comparing to BTC-per-one-BTC notional or per-share notional.
- Convert option premium to BTC-equivalent by scaling share option contract multiplier and BTC/share.
- Account for ETF fee, NAV premium/discount, and T-1 holdings data.

CME:

- Normalize from futures option to BTC spot-equivalent using futures price/basis.
- Include contract multiplier and whether using standard or micro BTC contract.
- Use settlement/open-interest stats from official CME feed/vendor.

OTC/RFQ:

- Normalize quote terms manually from term sheet.
- Include collateral and credit terms; these may dominate apparent vol difference.

3. Daily BTC Vol Desk Monitor — MVP format

Section 1: Market state

- BTC spot/reference
- IBIT last / NAV / premium-discount
- IBIT BTC/share
- CME front futures and basis
- Deribit futures/perps basis
- realized/historical vol reference

Section 2: Venue surface summary

- Deribit ATM IV by expiry
- IBIT ATM IV by expiry
- CME ATM IV by expiry
- 25-delta put IV by venue
- 25-delta call IV by venue
- risk reversal by venue
- term-structure slope by venue
- volume/OI by venue
- bid/ask quality by venue

Section 3: Dislocation board

For each candidate:

- Dislocation type: wrapper / skew / calendar / basis / margin / flow
- Instruments/legs
- Native prices
- BTC-equivalent normalized prices
- Gross edge
- Estimated costs: bid/ask, fees, margin, collateral, slippage
- Net edge
- Capacity estimate
- Execution confidence: screen-only / quote-verified / trade-verified
- Client application: treasury / miner / fund / family office / structured product
- Risk notes

Section 4: Client structure examples

- Treasury covered call quote grid
- Zero-cost collar grid
- Miner hedge ladder
- Fund relative-value spread package

Section 5: Missing data / warnings

- Source gaps
- stale fields
- wide markets
- unverified quote assumptions
- regulatory/venue constraints

4. Data gaps and vendor needs

Immediate gaps

- CME credentialed/production data source.
- OPRA-grade IBIT quote/trade history and live NBBO.
- Historical IBIT option IV/Greeks.
- Historical Deribit full surface data unless stored going forward or sourced from vendor.
- OTC/RFQ quote verification.
- Legal/compliance mapping for advisory/execution/principal activity.

Candidate vendors / infrastructure

- CME official WebSocket + DataMine.
- OPRA-licensed feeds.
- Databento for OPRA and possibly CME data depending license/product coverage.
- Cboe DataShop / LiveVol.
- OptionMetrics IvyDB US.
- ORATS.
- Polygon/Massive options APIs.
- dxFeed / Interactive Brokers / Bloomberg / Refinitiv / FactSet.
- Crypto options vendors for Deribit history/analytics if cheaper than self-storing.

5. Build sequence recommendation

Step 1: Deribit + IBIT daily snapshot

Build the first monitor using verified public/semi-official data:

- Deribit book summary + instruments.
- Nasdaq IBIT chain + Greeks.
- BlackRock iShares holdings CSV for BTC/share.
- OCC series search for official listed-series/OI cross-check.

Step 2: CME data-source decision

Choose one:

- Broker/vendor API for quick pilot.
- CME WebSocket/DataMine if institutional path and budget justify.
- Bloomberg/Refinitiv/manual export for investor memo proof only.

Step 3: Normalize and label confidence

Every result must show:

- source confidence

- execution confidence
- stale/as-of timestamp
- cost/margin assumptions

Step 4: Produce first daily report

Output should be suitable for two audiences:

- internal trader/structurer: exact instruments and dislocations
- investor/client: plain-English structures and why they exist

6. Current conclusion

The data spine is feasible, but the honest MVP must be labeled correctly:

- Deribit: usable immediately from official public API.
- IBIT: usable immediately for pilot via Nasdaq + BlackRock + OCC, but production-grade data needs OPRA/vendor.
- CME: official automated access exists, but requires credentials/licensing or vendor/broker access; public web pages must not be scraped.
- OTC/RFQ: must be quote-verified before any economics are presented as executable.

This supports the strategic thesis: the opportunity is not just IV comparison. The edge is normalizing different wrappers, margin regimes, collateral regimes, venue access constraints, and client flows into structures that a treasury, miner, fund, or allocator can actually use.